

Sem 1 2022-23 Mid Sem Exam, BITS Pilani KK Birla Goa Campus

Instructor in Charge: Harikrishnan N B

Student:

READ THE FOLLOWING CAREFULLY:

- **Make Up Exam**
 - **Honour Code for Students:** I shall be honest in my efforts and will make my parents proud. Write the oath and sign it on the answer sheet.
 - **Duration: 02:30 PM - 04:00 PM**
 - **Max. Marks = 30.**
 - **Students can use calculator. Mobile phones are strictly prohibited.**
 - **Answer all the questions.**
1. Decision Tree ID3 algorithm uses Shannon Entropy as a measure of impurity. Using Kullback–Leibler divergence mathematically show that Shannon Entropy is a measure of impurity. **6 marks**
 2. Consider the singular value decomposition (SVD) of the matrix $X = U\Sigma V^T$ of size $m \times n$. Answer the following.
 - (a) Using SVD how will you do Principal Component Analysis (PCA) of X . Write the steps. **2 marks**
 - (b) Consider the matrix transformation after PCA. Let $Y = X_{mean}P$, where X_{mean} is the mean subtracted matrix of X , P is the loadings. Mathematically show that the variance-covariance matrix of Y is a diagonal matrix. **6 marks**
 - (c) What does the diagonal entries of the variance-covariance matrix signifies? **2 marks**
 3. Answer the following questions:
 - (a) Write the mathematical formulation of the decision rule in logistic regression for binary classification problem? **2 marks**
 - (b) Write the mathematical formulation of the decision rule in perceptron algorithm for a binary classification problem? **2 marks**
 - (c) Write the mathematical formulation of the decision rule in Naive Bayes classifier for a binary classification problem? **2 marks**
 - (d) Given random variables X , Y , and Z . X is conditionally independent of Y given Z , denoted by $X \perp Y|Z$, if and only if: _____. **2 marks**
 4. Consider a binary classification problem. Define the following terms:
 - (a) Accuracy **1 mark**
 - (b) Precision **1 mark**
 - (c) Recall **1 mark**
 - (d) F1-score **1 mark**
 - (e) Why is accuracy not a good metric when it comes to imbalanced learning? Justify with an example. **2 marks**